ME403 – Introduction to Robotics

Lab 00: Introduction Week Handout

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1. Overview

This lab session focuses on setting up your development environment and verifying the connection with your assigned robotic arm. The lab supports two robots: the **Dobot Magician** (USB-serial) and the **DOBOT MG400** (TCP/IP over Ethernet). Scripts are organized into separate folders — `magician/` and `mg400/` — depending on which robot you are using.

Included Scripts

Script	Purpose
Dobot Magician (`magician/`)	
`01_init_check.py`	Verifies USB connection, clears alarms, and moves to safe ready pose. Run first.
`02_joint_control.py`	Interactive REPL: enter joint angles in degrees to move joints.
`03_relative_joint_control.py`	Body-frame FK exercise: enter relative joint angles, observe the conversion chain and FK prediction.
DOBOT MG400 (`mg400/`)	
`01_init_check.py`	Verifies Ethernet connection, enables robot, moves to READY_POSE. Run first.
`02_joint_control.py`	Interactive REPL: enter joint angles with FK preview and joint clamping.
`03_relative_joint_control.py`	Body-frame FK exercise: enter relative joint angles, observe the conversion chain and FK prediction.

2. Quick Start Guide

2.1. Dobot Magician (USB-serial)

Magician — Mamba (Recommended)

```
# 1. Activate environment and install
mamba activate dobot
pip install -r magician/requirements.txt

# 2. Verify connection
cd magician
python 01_init_check.py
```

Magician — Python venv

```
python3 -m venv .venv
source .venv/bin/activate     # Linux/macOS
# .venv\Scripts\activate      # Windows
pip install -r magician/requirements.txt
cd magician && python 01_init_check.py
```

2.2. DOBOT MG400 (Ethernet/TCP-IP)

MG400 — One-time SDK Setup

```
# Clone the official Dobot SDK into the vendor/ directory
cd dobot_ws
git clone https://github.com/Dobot-Arm/TCP-IP-4Axis-Python.git \
    vendor/TCP-IP-4Axis-Python
# Set PC Ethernet to static IP 192.168.2.100 / 255.255.255.0
# Verify: ping 192.168.2.9
```

MG400 — Run Scripts

```
pip install -r mg400/requirements.txt

# Default: Robot 1 (192.168.2.9)
cd mg400 && python 01_init_check.py

# Select a specific robot (1-4)
python 01_init_check.py --robot 2
```

3. Hardware Configuration

3.1. Dobot Magician (USB-serial)

1. **Surface:** Place on a flat, stable surface. Clear a 40 cm radius around the base.
2. **Posture:** Manually set the arm to a neutral posture (both segments at ≈45°) before powering on.
3. **Power:** Connect the 12V DC adapter. **USB power alone is insufficient.**
4. **Communication:** Connect the USB cable (CP210x USB-to-UART bridge).

Status Signals:

- **Green LED + Single Beep:** Ready.
- **Red LED + Rapid Beeping:** Alarm state — run `01_init_check.py` to clear.
- **Yellow/Flashing LED:** Busy (homing or executing a command).

3.2. DOBOT MG400 (Ethernet/TCP-IP)

1. **Surface:** Place on a flat, stable surface. Clear a 50 cm radius around the base (440 mm reach).
2. **Power:** Connect the DC power adapter. Wait for the status LED to turn solid green.
3. **Network:** Connect an Ethernet cable from the robot to your PC.
4. **PC Static IP:** Set your Ethernet adapter to `192.168.2.100`, netmask `255.255.255.0`.
5. **Verify:** `ping 192.168.2.9` — should respond with <1 ms latency.

Robot IP addresses (192.168.2.x subnet):

Robot	IP Address	CLI flag
1	192.168.2.9	`-robot 1` (default)
2	192.168.2.10	`-robot 2`
3	192.168.2.7	`-robot 3`
4	192.168.2.8	`-robot 4`

4. Software and OS Setup

4.1. Linux Setup (Dobot Magician only)

Users must have permission to access serial devices. Run the following once:

```
sudo usermod -a -G dialout $USER
# Log out and log back in for changes to take effect.
```

The Magician typically appears as `/dev/ttyUSB0` or `/dev/ttyACM0`. The MG400 uses Ethernet (no serial permission needed).

4.2. Windows Setup

The driver is usually handled by Windows Update. To find the port:

1. Open **Device Manager**.

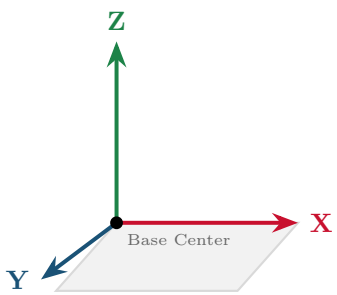
2. Expand **Ports (COM & LPT)**.
3. Look for **Silicon Labs CP210x USB to UART Bridge** (e.g., `COM3`).

4.3. macOS Setup

Modern macOS versions include the driver. The device will appear as: `/dev/cu.usbserial-*` or `/dev/cu.SLAB_USBtoUART`.

5. Dobot Magician Reference (USB-serial)

5.1. Coordinate System (Cartesian)



- **X-axis:** Forward (pointing away from base).
- **Y-axis:** Left/Right (Left is positive).
- **Z-axis:** Vertical (Up is positive).
- **Units:** Millimeters (mm).
- **Rotation (R):** Tool rotation in degrees.

5.2. Joint Definitions and Safe Limits

Joint	Motion	Safe Range	Units
J1	Base Rotation	±90°	deg
J2	Rear Arm Elevation	0° to 85°	deg
J3	Forearm Elevation	-10° to 85°	deg
J4	End-Effector Rotation	±90°	deg

Cartesian Dimension	Safe Operating Range	Unit
X (Reach)	120 to 315	mm
Y (Slew)	±158	mm
Z (Height)	5 to 155	mm

6. DOBOT MG400 Reference (Ethernet/TCP-IP)

6.1. Coordinate System

The MG400 uses the same X/Y/Z/R convention as the Magician, but with different operating bounds:

Axis	Safe Range	Unit	Note
X (Reach)	60 to 400	mm	Inner limit ≈60 mm (base singularity); max reach 440 mm
Y (Slew)	±220	mm	Symmetric full lateral sweep
Z (Height)	5 to 140	mm	**Z cannot go negative** — 0 mm = mounting surface
R (Rotation)	±170	deg	End-effector wrist rotation

READY_POSE = `(300, 0, 50, 0)` — safe home position above the surface.

6.2. Joint Definitions and Limits

Joint	Motion	Safe Range	Units
J1	Base Rotation	±170°	deg
J2	Shoulder Elevation	−5° to +90°	deg
J3	Elbow	−140° to +5°	deg
J4	Wrist Rotation	±170°	deg

7. Forward Kinematics Exercise (Script 03)

Script `03_relative_joint_control.py` introduces the concept of **body-frame (relative) joint angles** and makes the FK transformation chain visible in the terminal output.

7.1. What Are Relative Joint Angles?

Body-Frame vs. World-Frame Angles

Each joint angle can be measured from two different reference directions:

Joint	Relative (body-frame)	Absolute (world-frame)
J1	Base rotation from world X-axis	Same as relative
J2	Shoulder elevation from horizontal	Same as relative
J3	Elbow offset *from the upper arm*	$= \mathrm{J2} + \mathrm{J3}_{\mathrm{rel}}$
J4	Wrist offset *from the forearm*	$= \mathrm{J3}_{\mathrm{abs}} + \mathrm{J4}_{\mathrm{rel}}$

J1 and J2 are identical in both conventions. J3 and J4 are where they differ: a relative J3 of 10° means the forearm is 10° above the upper arm, regardless of how high the shoulder is.

7.2. Conversion Chain

The script applies this chain before every move:

$$j_{3,\text{abs}} = j_{2,\text{rel}} + j_{3,\text{rel}}$$
$$j_{4,\text{abs}} = j_{3,\text{abs}} + j_{4,\text{rel}} = j_{2,\text{rel}} + j_{3,\text{rel}} + j_{4,\text{rel}}$$

Magician firmware note: the Dobot Magician uses a parallel linkage and its firmware already interprets the J3 channel as a body-frame offset (so $j_{3,\text{fw}} = j_{3,\text{rel}}$ trivially). Only J4 needs the absolute wrist angle: $j_{4,\text{fw}} = j_{4,\text{abs}}$.

MG400 firmware note: the MG400 firmware expects fully absolute angles, so both J3 and J4 must be accumulated: $j_{3,\text{fw}} = j_{3,\text{abs}}$ and $j_{4,\text{fw}} = j_{4,\text{abs}}$.

7.3. Worked Example (Dobot Magician)

Input: `0 20 10 0` (all relative)

Step	Value
$j_{3,\text{abs}}$	$= 20° + 10° = 30°$
$j_{4,\text{abs}}$	$= 30° + 0° = 30°$
$j_{3,\text{fw}}$	$= 10°$ (trivial for Magician)
$j_{4,\text{fw}}$	$= 30°$ $(= j_{4,\text{abs}})$
FK reach	$= 135\cos(20°) + 147\cos(30°) \approx 254\,\text{mm}$
FK height	$= 135\sin(20°) + 147\sin(30°) \approx 120\,\text{mm}$

Safety Note

Joint bounds are always applied to the **firmware angles** (absolute), not the relative inputs. Entering $j_{3,\text{rel}} = 10°$ when $j_{2,\text{rel}} = 80°$ would produce $j_{3,\text{abs}} = 90°$, which exceeds the J3 firmware bound of 85° and will be clamped with a warning before the move executes.

7.4. Usage and Terminal Output

```
# Dobot Magician
cd magician && python 03_relative_joint_control.py

# DOBOT MG400
cd mg400 && python 03_relative_joint_control.py
python 03_relative_joint_control.py --robot 2
```

Sample terminal output for input `0 20 10 0`:

```
Abs: J1=  0.00  J2= 20.00  J3= 30.00  J4= 30.00
Rel: J1=  0.00  J2= 20.00  J3= 10.00  J4=  0.00
> 0 20 10 0

  Relative:  J1_rel=   0.00  J2_rel=  20.00  J3_rel=  10.00  J4_rel=   0.00
  Absolute:  j1_abs=   0.00  j2_abs=  20.00  j3_abs=  30.00  j4_abs=  30.00
  Firmware:  j1_fw=    0.00  j2_fw=   20.00  j3_fw=   10.00  j4_fw=   30.00
  [FK]       X= 254.17  Y=   0.00  Z= 119.72  R= 30.00
  [Move]     J1=  0.00  J2= 20.00  J3= 10.00  J4= 30.00
  Done.      J1=  0.00  J2= 20.00  J3= 10.00  J4= 30.00
```

```
            X= 253.90   Y=    0.00   Z= 119.50
```

Challenge: before running the script, calculate the expected X, Y, Z by hand using the FK formulas. Then compare your answer with the `[FK]` line in the terminal.

8. Troubleshooting

8.1. Dobot Magician

Magician: Common Connection Issues

Symptom	Recommended Action
`Access Denied` or `Port Busy`	Ensure **DobotStudio** and all other Python scripts are closed. Only one process can own the serial port.
`Device not found`	Verify both USB and Power cables. Check Device Manager (Win) or `ls /dev/tty*` (Unix).
Motors don't move	Check the 12V power supply. If the light is red, run `01_init_check.py` to clear alarms.
Erratic movement	Reset to neutral posture and power cycle the robot.

8.2. DOBOT MG400

MG400: Common Issues

Symptom	Recommended Action
`dobot_api.py not found`	Clone the SDK: `git clone .../TCP-IP-4Axis-Python.git vendor/TCP-IP-4Axis-Python`
`Connection refused`	Check Ethernet cable, verify PC IP is 192.168.2.100, ping the robot IP.
Robot in ERROR mode	Run `check_errors(dashboard)` — if uncleared, press E-stop, release, power-cycle.
Z motion stops early	MG400 Z cannot go negative. Ensure all targets have $Z \geq 5\,\text{mm}$.

9. Safety Mandates

Mandatory Safety Rules (Both Robots)

1. **Workspace:** Maintain a clean area. No cables or personal items in the reach zone (40 cm for Magician; 50 cm for MG400).
2. **Initialization:** Always run `01_init_check.py` before starting your own scripts.
3. **Helper Functions:** Use `safe_move()` from `utils.py` / `utils_mg400.py`. It enforces software limits to prevent physical damage.
4. **Supervision:** Stay within reach of the power cable (or E-stop for MG400) for immediate emergency disconnection.
5. **MG400 only:** Z cannot go negative ($Z \geq 5$ mm). Never command the tool below the mounting surface.